

Abstract

Agriculture is one of the most important sectors for food production, but farmers face numerous challenges such as labor shortages, crop diseases, weed growth, pest attacks, water scarcity, animal intrusion, and increasing production costs. Traditional farming methods often rely on manual monitoring and excessive use of water, fertilizers, and pesticides, leading to reduced productivity and environmental concerns. To address these issues, this project proposes an AI-Powered Smart Agricultural Robot for Precision Farming with Robotic Harvesting, Blockchain Crop Traceability, and a Self-Learning AI Recommendation System. The proposed system integrates Artificial Intelligence (AI), Internet of Things (IoT), Robotics, Computer Vision, Blockchain, and Mobile Application technologies into a single autonomous agricultural platform. The robot uses an AI camera and environmental sensors to continuously monitor crop health, soil moisture, temperature, humidity, water level, and field conditions. AI algorithms detect crop diseases, weeds, pests, nutrient deficiencies, fruit ripeness, and animal intrusion in real time. Based on the analysis, the system recommends both organic and chemical treatments through a mobile application. If the farmer selects the organic option, the application provides preparation instructions before the robot performs precision spraying. Smart irrigation is achieved by monitoring soil moisture and activating the water pump only after farmer approval or in automatic mode. The robot also incorporates robotic harvesting to collect ripe crops identified by AI-based ripeness detection. To improve transparency and food safety, blockchain technology records crop information, including cultivation history, treatments, harvest date, and GPS location, while QR codes enable consumers to verify product authenticity. A unique feature of the proposed system is the Self-Learning AI Recommendation System, which continuously improves treatment recommendations by learning from farmer feedback and comparing before-and-after crop conditions. This enables the robot to provide increasingly accurate and personalized recommendations over time. Powered by solar energy and rechargeable batteries, the proposed robot offers an environmentally friendly and cost-effective solution for modern precision agriculture.

Keywords: Artificial Intelligence (AI), Internet of Things (IoT), Precision Agriculture, Smart Agricultural Robot, Computer Vision, Crop Disease Detection, Weed Detection, Pest Detection, Smart Irrigation, Precision Spraying, Robotic Harvesting, Blockchain.

